

**CAR #17 arrives 8:00 AM • battery 25% • needs ~80% by 5:00 PM
parked 9 h but needs only ~3 h of charging → plenty of slack, NOT urgent**

Charging company's plan: "Wait now, I'll charge you midday
when the lot is quiet and the grid is calm (urgent cars go first)."

9:00 AM — the driver decides:

FOLLOWS the plan

waits • charges midday
leaves at 82% full

Driver ✓

Company ✓

Grid ✓

**ALL THREE HAPPY
DAY PASSES ✓**

**IGNORES plan — grabs a
charger now, out of turn**

bumps urgent Car #4 → #4 misses its noon trip

everyone charges at once → grid peak spikes

out-of-turn requests → queue jams, long waits

Driver ✗

Company ✗

Grid ✗

**Car #17 itself leaves fine (85%)
but the DAY FAILS ✗**

**Only about 1 driver in 20 does this — but with dozens of cars, almost every day gets one.
One selfish grab, in one hour, flips the whole day. That is why "all three happy"
crashes from ~97% of days to ~1% of days — and why the fix is a small, targeted reward to wait.**